Image Resampling

Image Scaling

This image is too big to fit on the screen. How can we generate a half-sized version?

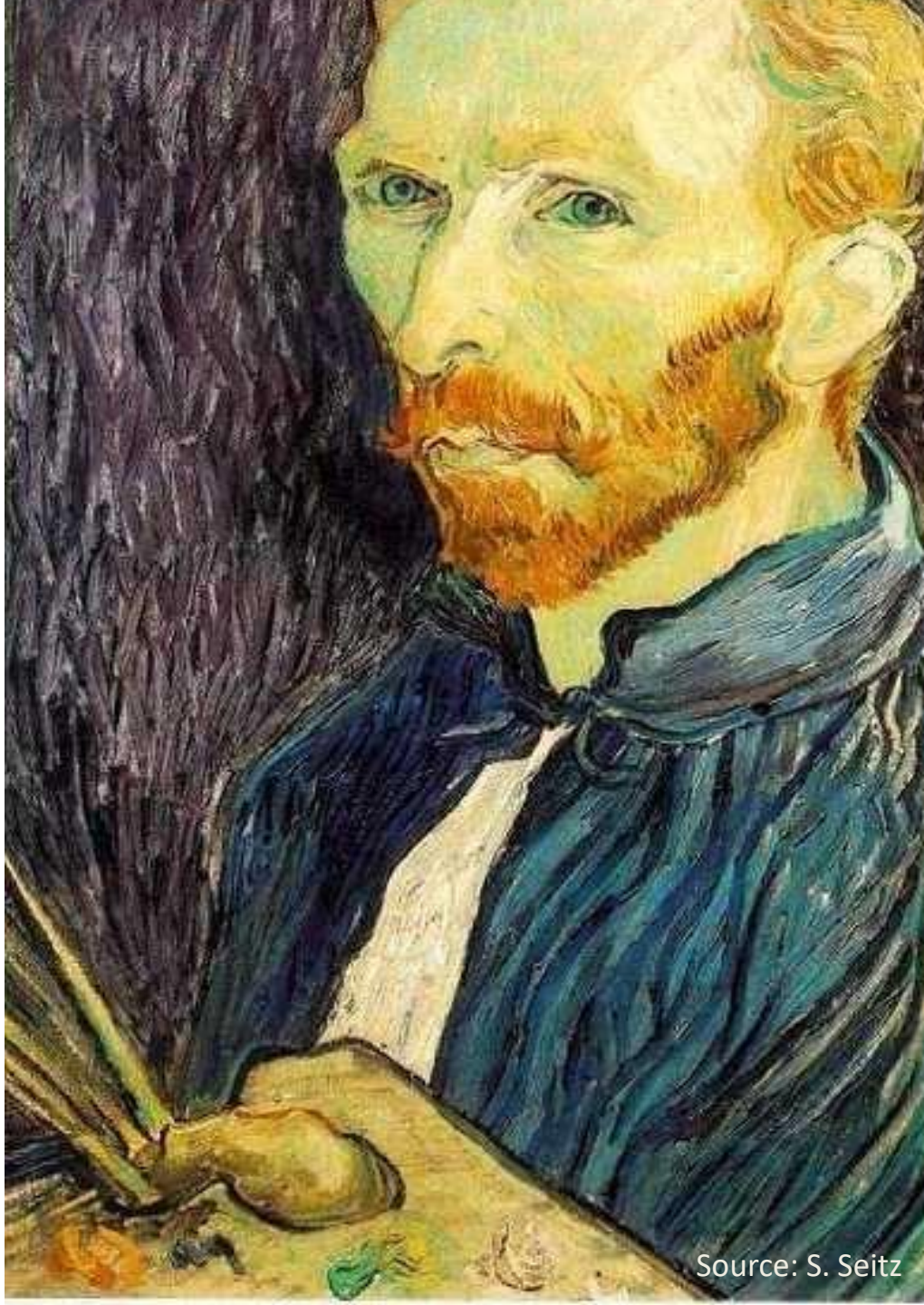
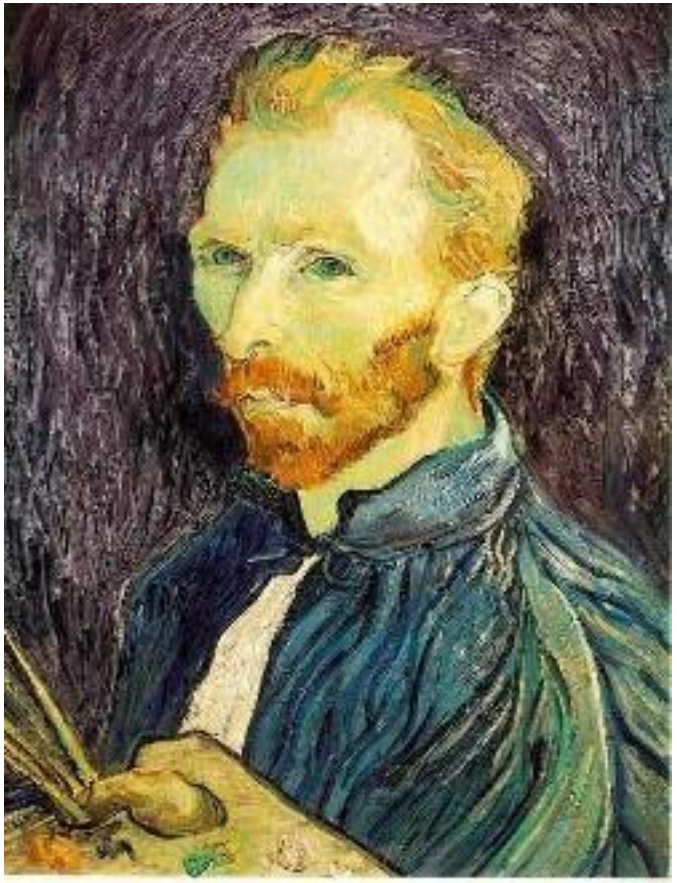


Image sub-sampling



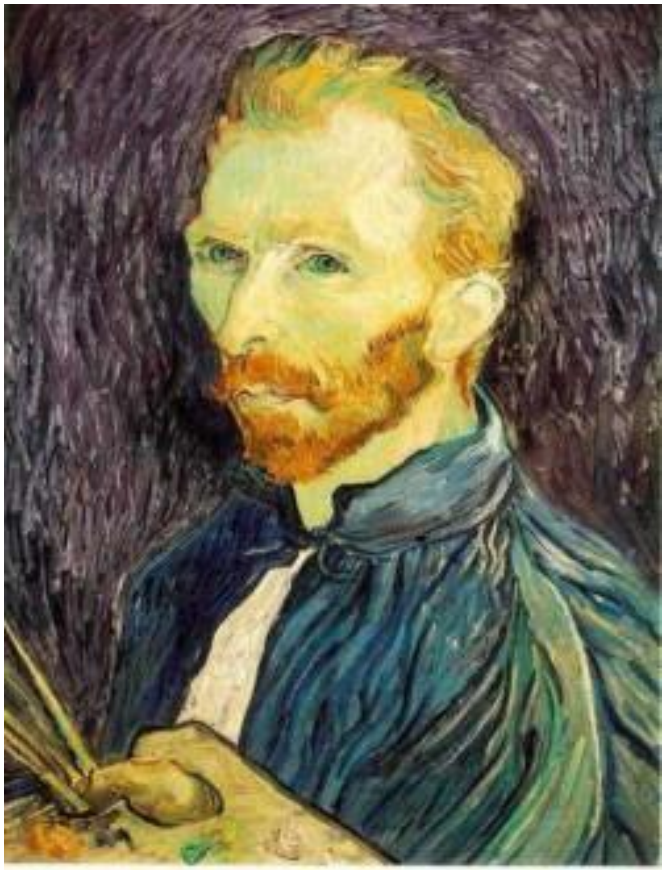
1/4

1/8

Throw away every other row and column to create a smaller size image
- called *image sub-sampling*

Source: S. Seitz

Gaussian pre-filtering



Gaussian 1/2

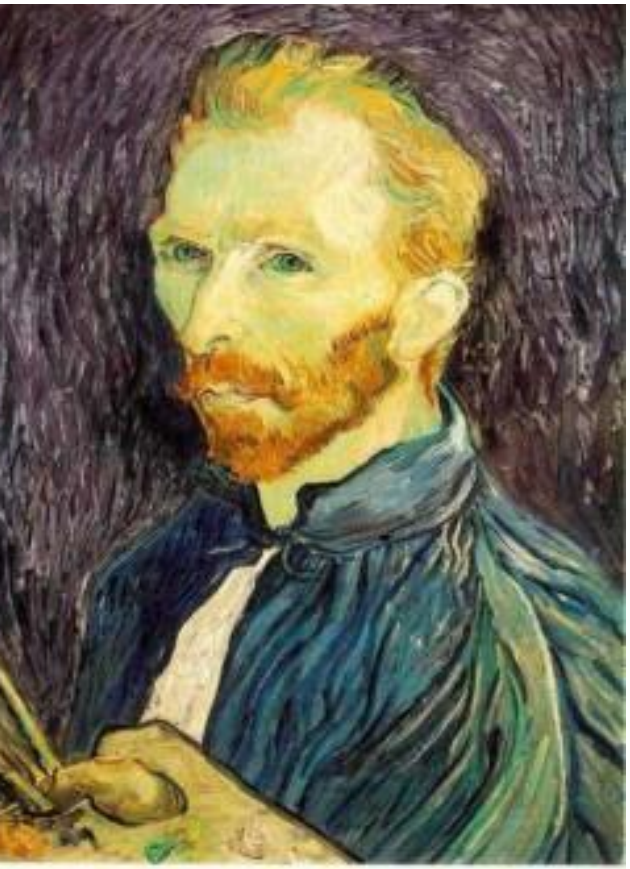
G 1/4

G 1/8

- Solution: filter the image, *then* subsample

Source: S. Seitz

Subsampling with Gaussian pre-filtering



Gaussian 1/2

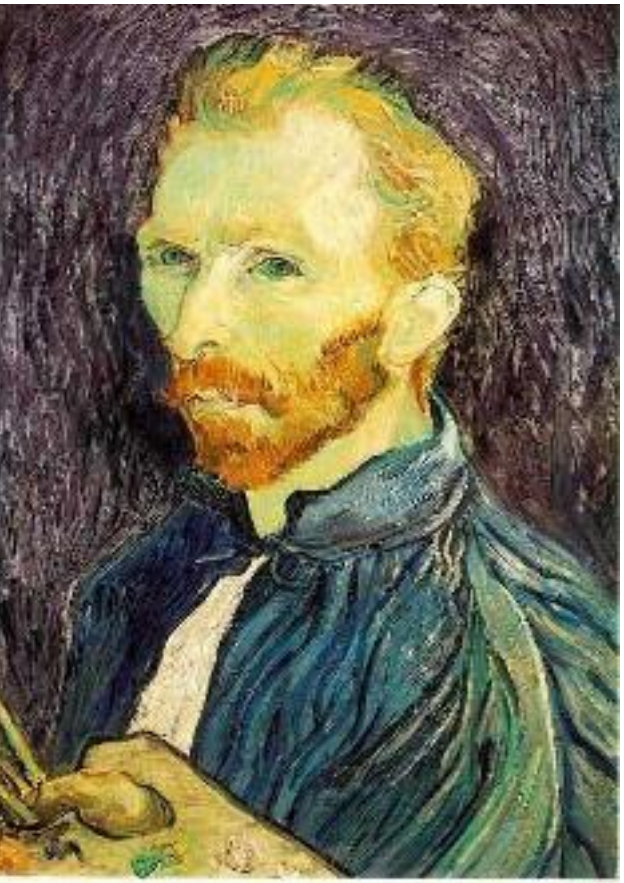
G 1/4

G 1/8

- Solution: filter the image, *then* subsample

Source: S. Seitz

Compare with...



1/2

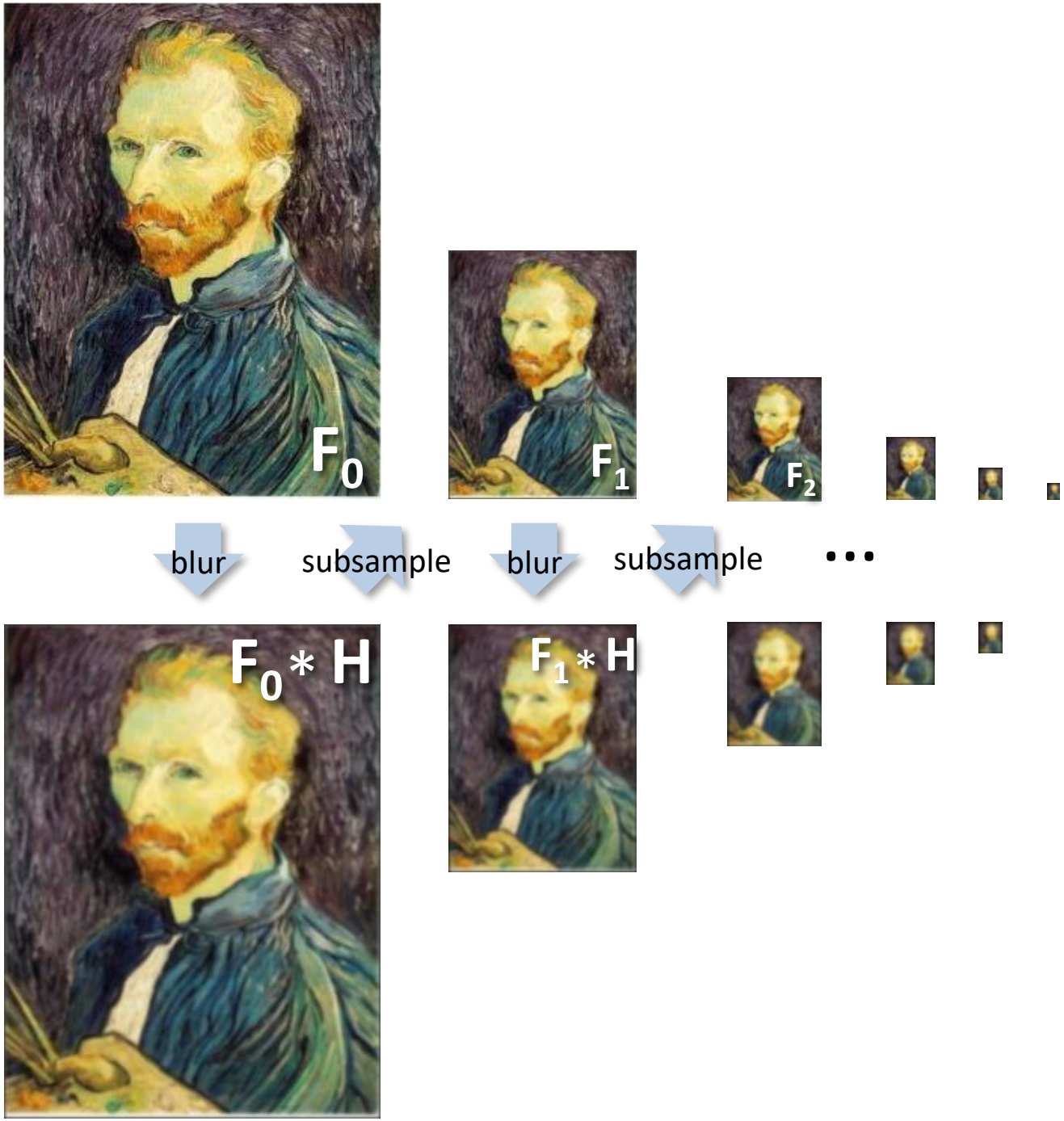
1/4 (2x zoom)

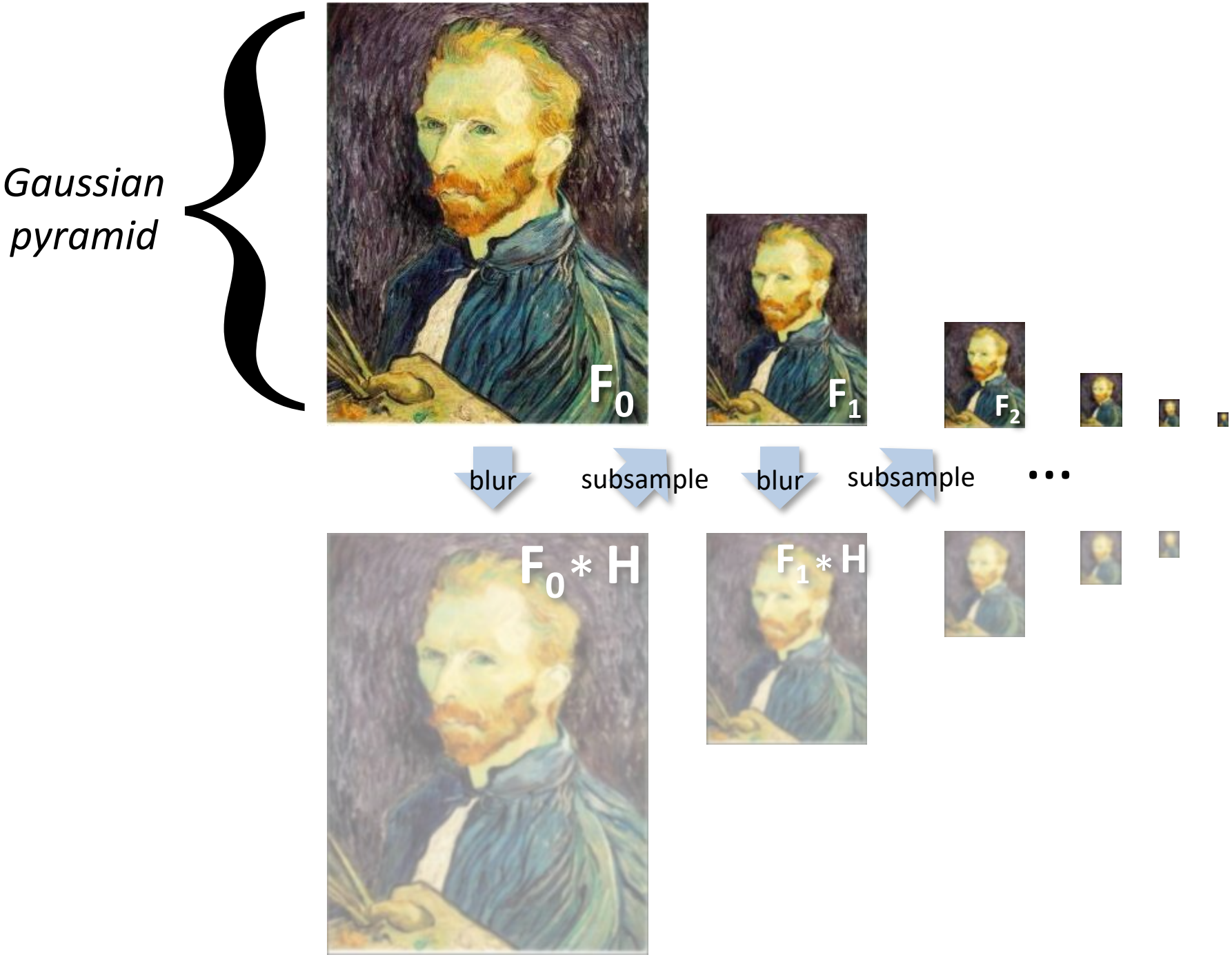
1/8 (4x zoom)

Source: S. Seitz

Gaussian pre-filtering

- Solution: filter the image, *then* subsample




Gaussian pyramid
F_0
F_1
F_2
blur
subsample
blur
subsample
...
$F_0 * H$
$F_1 * H$

Upsampling

- This image is too small for this screen:
- How can we make it 10 times as big?
- Simplest approach:

  repeat each row

  and column 10 times
- (“Nearest neighbor

  interpolation”)